

Railway Reservation System

Main Use Case Flow

Project Title: Railway Reservation System with Booking and Cancellation Features

1. Actors

Actor	Main Goals
Passenger / User	Register, login, search trains, check seat availability, book tickets, make payment, view bookings, cancel tickets, and check refund status.
Administrator	Login, manage trains, stations, routes, schedules, seats, bookings, and cancellations.
Payment Gateway	Process ticket payments and refunds and return payment status.

2. Passenger – Main Booking Flow

Passenger → Register / Login → Search Train → Check Seat Availability → Select Train & Class → Enter Passenger Details → Calculate Fare → Make Payment → Payment Successful? → Confirm Ticket → Generate PNR → View Ticket

If seats are unavailable: Check Seat Availability → No → Display “No Seats Available” → End.

If payment fails: Make Payment → No → Display “Payment Failed” → Booking is not confirmed.

3. Ticket Cancellation Flow

Passenger → Login → View My Booking → Select PNR → Verify Booking → Valid Booking? → Click Cancel → Calculate Refund / Cancellation Charge → Confirm Cancellation → Update Booking Status to Cancelled → Release Seat → Process Refund → Display Cancellation Confirmation

If the PNR is invalid: Verify Booking → No → Display “Invalid Booking / PNR” → End.

4. Administrator – Main Flow

Administrator → Admin Login → Admin Dashboard → Manage Trains / Manage Stations / Manage Routes / Manage Schedules / Manage Seats / View Bookings / View Cancellations

5. Main Use Cases

Passenger	Register; Login; Search Train; Check Seat Availability; Book Ticket; Make Payment; View Booking; Cancel Ticket; View Refund Status
Administrator	Login; Manage Trains; Manage Stations; Manage Routes; Manage Schedules; Manage Seats; View Bookings; View Cancellations
Payment Gateway	Process Payment; Process Refund

6. UML Include Relationships

Book Ticket <<include>> Check Seat Availability, Enter Passenger Details, Calculate Fare, Make Payment, Generate PNR.

Cancel Ticket <<include>> Verify Booking, Calculate Refund, Process Refund, Update Seat Availability.

Note: In the UML Use Case Diagram, Passenger, Administrator, and Payment Gateway are external actors. The Railway Reservation System is the system boundary, not an actor.